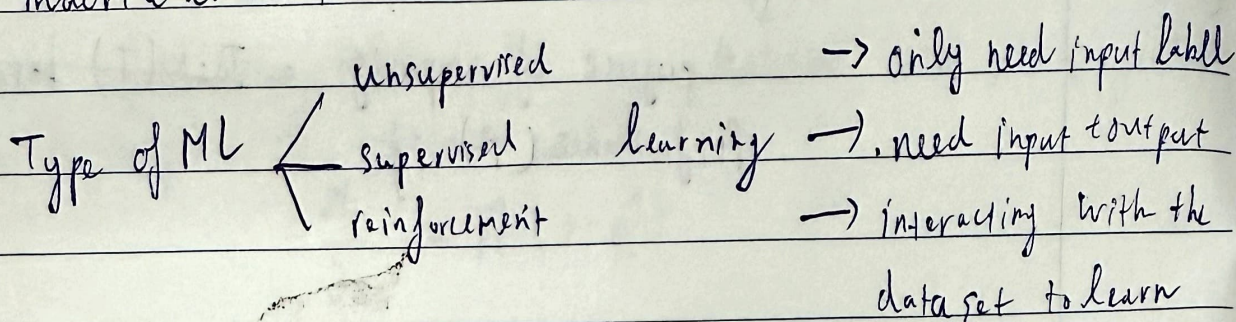


* ML def: if a Performance of some tasks (?) is improved, the machine learns it.



Classification: used for discrete dataset

Regression: used for continuous dataset

* Pipeline

Data → features → model → train → evaluation → prediction
→ deployment

If the process is not good, we go back to training phase or re design the model.

Metric

$$\text{Accuracy} = \frac{TP + TN}{\text{Total cases}}$$

$$\text{Precision} = \frac{TP}{TP + FN}$$

$$F_1 = H(\text{Precision}, \text{Recall})$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

where $H()$ is Harmonic Mean

$$(H(m, n)) = \frac{2 \cdot m \cdot n}{m + n}$$

* Evaluation

Underfit : fails to capture the pattern

Overfit : low training errors but high test error

Good fit : Enough for both training and test errors

* Bias and Variance

Bias : Error from underfit

Variance : Error from overfit

Our goal is to find trade-off between Bias and Variance.

* Some techniques should be used to read graph

When reading graph, we expect $F1$ high and MSE low.

← we can check error points by looking at the y-axis

What to expect from ML

← we want to generalize data, not memorize data

← we want to make a model that can balance bias and variance

Some notes on Evaluation

Use k-cross to try and find best model for that specific ~~model~~ dataset

Bootstrap is to ^{test} ~~test~~ if the model handles with the repeated data good enough

Decision Tree

Def: Basically we use tree in light of a sequence of question to find out the category that one ^{data} set belongs to.

Some formula used for ~~decide~~ this tree

$$\text{Entropy} = - \sum_{i=1}^k p_i \log_2 p_i$$

This is used to know the uncertainty of Dataset
* Read the examples in the slide

In gain $\rightarrow$ some formula

$$\text{Gini}(S) = 1 - \sum_{i=1}^k p_i^2$$

Please remember that there is $\log(0)$ in Entropy and in gain as they mean the data set is pure.

* Continuous Attribute

Like the name suggests, it's an attribute that accepts a continuous range of value.

Eg. Temp is splitted into two categories

$t < 30$ or and $t \geq 30$ categories

* Missing Value

The attribute of dataset is not recognized due to some error

Some ways to fix it is to use Imputation or CART

Stopping and pruning

Stopping:

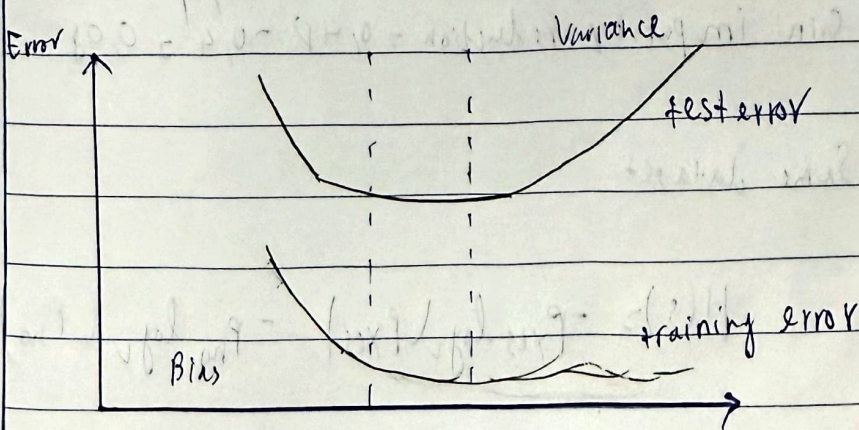
At some conditions we cut $\rightarrow$ stop the tree

Pruning: used to prevent overfit

Pre-pruning
Post-pruning

& Note Pruning too much can cause underfit

Examples for ML intro



model complexity

source: slide 27 - Introduction to ML

In the LHS, both the errors are too high which means underfitting

- When increasing model complexity, the errors continue to ~~increase~~ decrease and then increase back

We should choose the region in between two dotted lines via validation usage to get the right model complexity

Tree examples

10 samples: 6 yes, 4 no. Left branch has 4 yes, 1 No, right branch has 2 yes, 3 no.

$$\text{Parent Gini impurity} = 1 - 0,6^2 - 0,4^2 = 0,48$$

$$\text{LC Gini impurity} = 1 - \left(\frac{4}{5}\right)^2 - \left(\frac{1}{5}\right)^2 = 0,32$$

$$\text{RC} = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0,48$$

$$\text{WGini improved after split} = 0,5 \times 0,32 + 0,5 \times 0,48$$

$$= 0,4$$

$$\text{Gini impurity reduction} = 0,48 - 0,4 = 0,08$$

Same dataset

$$H(s) = -p_{\text{yes}} \log_2(p_{\text{yes}}) - p_{\text{no}} \log_2(p_{\text{no}})$$

$$H_{\text{parent}} = -0,6 \log_2(0,6) - 0,4 \log_2(0,4) \approx 0,971$$

$$H_{\text{LC}} = -0,8 \log_2(0,8) - 0,2 \log_2(0,2) \approx 0,722$$

$$H_{\text{RC}} = -0,4 \log_2(0,4) - 0,6 \log_2(0,6) \approx 0,971$$

$$IG = 0,971 - \frac{5}{10} H_{\text{LC}} - \frac{5}{10} H_{\text{RC}} \approx 0,125$$